CARBOPLATIN

Dose by AUC (Calvert)
Myelosuppression (platelets)

CHEMICAL NAME cis-diammine(cyclobutane-1,1-dicarboxylato)platinum(II)
CLASS Platinum-based cytotoxic agent
TRADE NAMES Paraplatin · Carboplatin injection · Carbosin

THERAPEUTIC USES / INDICATIONS

- Ovarian cancer (with paclitaxel).
- Lung cancer (NSCLC, SCLC), head and neck, testicular seminoma, bladder, endometrial cancers.
- Children's tumours; alternative to cisplatin in renal impairment/hearing loss.

ADMINISTRATION

- **IV** infusion over 15–60 min in glucose 5% or saline.
- No aggressive hydration needed (less nephrotoxic).
- Avoid aluminium needles.
- Hypersensitivity risk ↑ after several cycles (≥6).

DOSAGES

- Calvert formula: dose (mg) = target AUC × (GFR + 25).
- **Typical AUC:** 5–6 (with paclitaxel) every 3 weeks; AUC 2 weekly.
- GFR measured or estimated (capped).

PHARMACOKINETICS

- Less protein binding than cisplatin.
- Excreted mainly in urine (dose depends on renal function). **t½:** 1.5–6 h (free platinum).

PHARMACODYNAMICS

- Forms **platinum–DNA cross-links** (slower aquation than cisplatin) → apoptosis.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting (moderate).
- Fatigue.
- Mild alopecia.

ADVERSE EFFECTS (SERIOUS)

- Myelosuppression (thrombocytopenia dose-limiting, nadir ~day 21; anaemia).
- Hypersensitivity/anaphylaxis (especially after multiple cycles).
- Neuropathy, ototoxicity, nephrotoxicity (less than cisplatin).
- Electrolyte loss (Mg^{2+}).

CONTRAINDICATIONS & CAUTIONS

- Severe myelosuppression; severe renal impairment (CrCl <20 without adjusted dosing).
- Significant bleeding; previous platinum anaphylaxis.
- Pregnancy, breastfeeding.

KEY INTERACTIONS

- **Aminoglycosides:** ↑ nephro/ototoxicity.
- **Myelosuppressive drugs:** additive.
- **Live vaccines:** contraindicated.

NURSING MANAGEMENT

- Check FBC (platelets, neutrophils), renal function before each cycle.
- Monitor closely for hypersensitivity, especially later cycles; anaphylaxis kit ready.
- Bleeding and infection precautions.
- Antiemetics as per protocol.

PATIENT EDUCATION

- Report bruising, bleeding, fever ≥38 °C.
- Report itching, flushing or breathlessness during infusion.
- Fatigue is common; rest and gradual activity.

KEY POINTS

- Platinum agent dosed by AUC (Calvert).
- Less nephrotoxic/emetogenic than cisplatin.
- Thrombocytopenia dose-limiting.
- Hypersensitivity after many cycles.